

1961



Soviet pilot Yuri Gagarin was selected from 20 candidates to be the first man in space.

IN FEBRUARY, PHYSICISTS at the University of California Berkeley succeeded in producing **atoms of a new heavy element** by bombarding a sample of the element californium with nuclei of boron atoms. They called the element **lawrencium**, after American physicist **Ernest Lawrence**, the inventor of the cyclotron—a particle accelerator. With an atomic number of 103, lawrencium was the last—and the heaviest—of a group of radioactive metals called actinides.

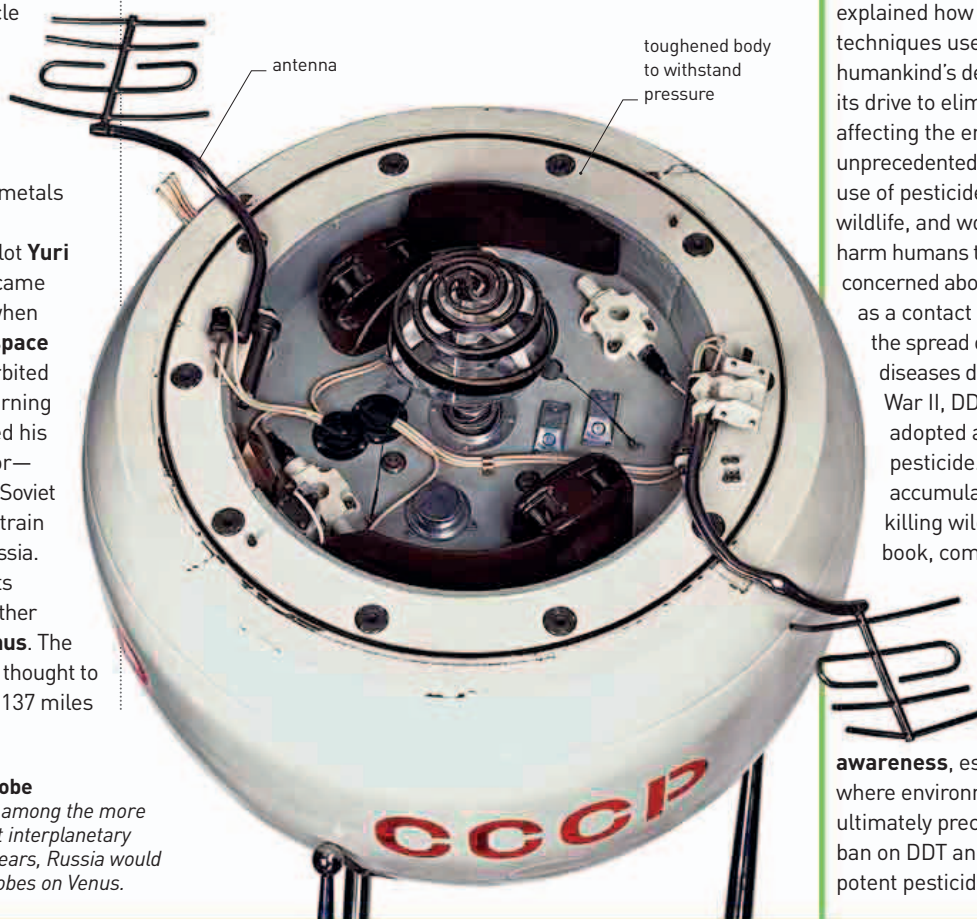
On April 12, Soviet pilot **Yuri Gagarin** (1934–68) became the **first cosmonaut** when he **traveled to outer space** aboard Vostok 1. He orbited Earth once before returning safely and was awarded his country's highest honor—the title of "Hero of the Soviet Union." He went on to train new cosmonauts in Russia.

The USSR launched its **Venera program** to gather **information about Venus**. The first probe, Venera 1, is thought to have passed within 62,137 miles

(100,000 km) of Venus. It was the first man-made object to fly by another planet.

In April, prompted by the plight of wildlife in Africa and elsewhere, a team including biologist **Julian Huxley** (1887–1975) and ornithologist **Peter Scott** (1909–89) proposed the establishment of an international

organization for wildlife conservation. They created the International Secretariat of the **World Wildlife Fund (WWF)**—now known as the World Wide Fund for Nature—in Switzerland. The WWF went on to set up offices worldwide and harnessed the expertise of scientists to protect endangered species.



Soviet Venera space probe

The Venera probes were among the more sophisticated of the first interplanetary space probes. Over the years, Russia would succeed in landing 10 probes on Venus.

1962



DDT was developed as a contact poison to control insect pests and was claimed to be harmless to people and the environment.

IN JUNE, *The New Yorker* magazine began to serialize a book by American marine biologist **Rachel Carson**, titled ***Silent Spring***. It proclaimed that human activity, particularly the **use of pesticides** such as **DDT** (dichlorodiphenyltrichloroethane), threatened the environment with damage and destruction. Carson explained how the intensive techniques used to satisfy humankind's demand for food and its drive to eliminate pests were affecting the environment on an unprecedented scale. Widespread use of pesticides was harming wildlife, and would ultimately harm humans too. Carson was concerned about DDT. Developed as a contact poison to control the spread of insect-borne diseases during World War II, DDT was later adopted as an agricultural pesticide. However, it accumulated in food chains, killing wildlife. Carson's book, coming a year after the inauguration of the WWF, served as an alarm call. It created a **new environmental awareness**, especially in the US, where environmental concerns ultimately precipitated a national ban on DDT and other highly potent pesticides.

In July, the multinational **communications satellite Telstar** was sent into space aboard a NASA rocket. The first of its kind, Telstar made it possible for **live television signals** to be transmitted across the Atlantic. The first pictures were seen on television screens on July 10. Telstar became the prototype for later, more efficient, models.



RACHEL CARSON
(1907–64)

Trained in marine biology, Rachel Carson achieved acclaim as a writer of popular natural history books before she became famous for her book, *Silent Spring*. Her work led to the establishment of the US Environmental Protection Agency. She was posthumously awarded the Presidential Medal of Freedom in 1980.

February 14
Lawrencium is
synthesized

April 12 Soviet
pilot Yuri Gagarin,
the first human in
space, orbits Earth

April 29
World Wildlife Fund
(WWF) founded

May 25 The US Congress
announces launch of
Apollo program
to land a man on
the Moon

May 19 Venera 1
passes close
to Venus

November 9
First edition of
International Code of
Zoological Nomenclature
is designed to standardize
naming of animal species

February 20
John Glenn
becomes first
American to
orbit Earth

June The New
Yorker begins
serialization of
Rachel Carson's
Silent Spring

July 10 Telstar,
the first active
communications
satellite, is launched